

1

B

11

C



D

E

**F**

Id: 1/3

11

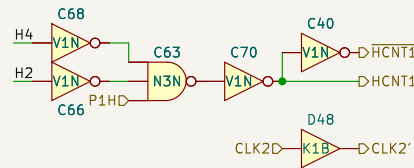
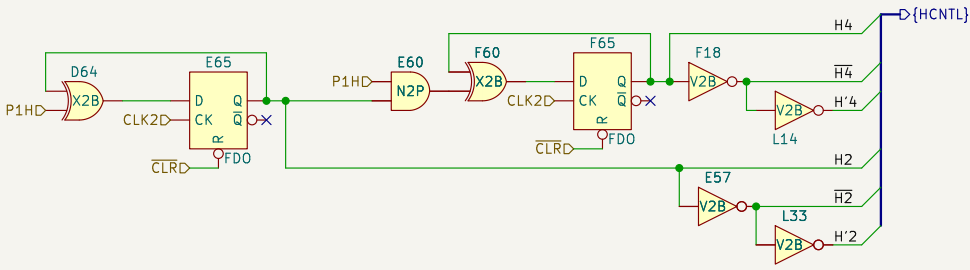
11

1

Horizontal clock P1H and  $\overline{\text{CLR}}$  initiates the timing of the 007780. H2 and H4 are also generated which creates a 3-bit counter, HCNT, which controls the timing of the device. There are  $2^3 = 8$  different counter values: 0..7.

HCNT 0,1 Render Tilemap 1  
HCNT 2,3 Render Tilemap 2  
HCNT 4,5 Idle, writing to page 0  
HCNT 6,7 CPU read/write cycle

TIM2 is active low during HCNT cycles 6 and 7.  
When HCNT is 4 or 5 then row address = 0. The column address is incremented by one for each HCNT cycle. Is this used to not wear out the DRAM modules?



### DRAM Control Signals

